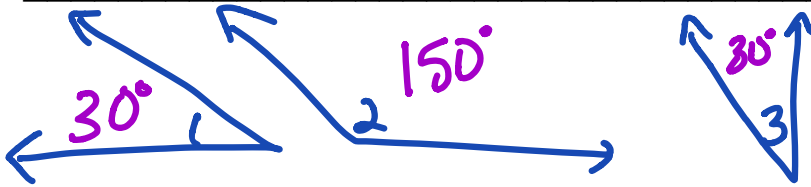
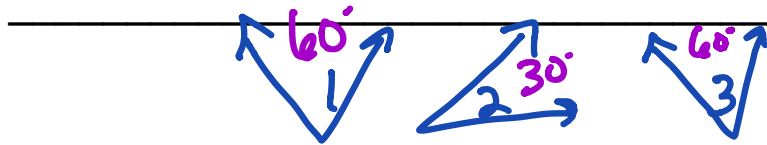


Congruent Supplements Theorem

If $\angle 1$ and $\angle 2$ are supplementary, and $\angle 2$ and $\angle 3$ are supplementary, then $\angle 1 \cong \angle 3$.

**Congruent Complements Theorem**

If $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary, then $\angle 1 \cong \angle 3$.

**Linear Pair Postulate**

If 2 angles form a line, then they are supplementary.
 $\angle 1$ and $\angle 2$ are a linear pair $\rightarrow$ given supplementary.
 $\angle 1$ and $\angle 2$ are supplementary $\rightarrow$ Linear Pair Post.

Vertical Angle Theorem

Vertical angles are ALWAYS = or $\cong$

Example: Solve for the variable.

a)

$$9y + 7 = 2y + 98$$

$$y = 13$$

b)

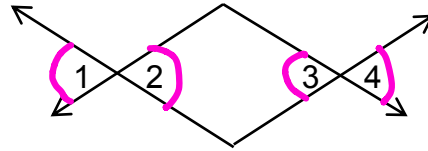
$$3x + 8 = 6x - 22$$

$$x = 10$$

Practice Proof

Given: $\angle 2 \cong \angle 3$

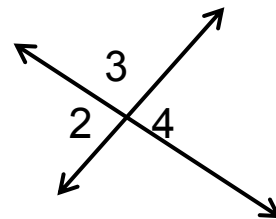
Prove: $\angle 1 \cong \angle 4$



Statements	Reasons
① $\angle 2 \cong \angle 3$	① given
② $\angle 1 \cong \angle 2$ $\angle 3 \cong \angle 4$	② Vertical $\angle$ Thm
③ $\angle 1 \cong \angle 3$	③ Transitive Prop $\cong$
④ $\angle 1 \cong \angle 4$	④ Transitive Prop. of $\cong$

Given: $\angle 2$ and $\angle 4$ are vertical angles
 $\angle 2$ and $\angle 3$ are a linear pair
 $\angle 3$ and $\angle 4$ are a linear pair

Prove: $\angle 2 \cong \angle 4$



Statements	Reasons
① $\angle 2$ and $\angle 3$ are a linear pair $\angle 3$ and $\angle 4$ are a linear pair	① given
② $\angle 2$ and $\angle 3$ are supplen. $\angle 3$ and $\angle 4$ are supplen.	② Linear Pair Post.
③ $\angle 2 \cong \angle 4$	③ Congruent Supplements Thm

